

Exploring the difference in volume to surface area ratio
between the volumes of revolutions generated by various
aeronautical bodies

1. Introduction

As part of my work experience this summer, I worked at an engineering firm where I was exposed to modelling software used in the industry. One of my tasks at this firm involved modelling an airfoil within this software. In my research for this task, I came across the Sears-Haack body, a shape that minimises wave drag in supersonic flight. (*Sears-Haack Body*) Inspired by this, I aimed to compare similar objects by creating volumes of revolutions from one of my own airfoils designed using modelling software Rhino8 (Robert McNeel & Associates) and Grasshopper, software within Rhino8, to see how the volume to surface area ratio of these objects compare to the Sears-Haack body. The NACA0012 airfoil has been chosen as an industrial airfoil design specifically due to its symmetrical nature, as solids of revolution require axial symmetry in order to be generated. (NACA 0012 AIRFOILS (n0012-II)) I will be analysing these modelled objects through surface area and volume of revolutions and compare the results I obtain from each to see how these physical properties affect the design and aerodynamics of the objects. Despite the fact that airfoils are 2D shapes that represent the cross section of an aircraft wing, I found them to be an appropriate shape to be modelled in 3D due to their aeronautical context and the general principles behind their design, that is to reduce drag and produce lift during flight.

Hypothesis

I hypothesise that the Sears-Haack body will have the lowest surface area to volume ratio of all 3 objects as based on my physics knowledge the object with the lowest surface area will have the least drag force acting on it. Since the Sears-Haack body minimises this drag, it should also have the lowest surface area. Next, I hypothesise that the real NACA0012 airfoil will have the second lowest surface area to volume ratio as it is a real industrial airfoil that will have been carefully designed to make it more aerodynamic. Finally, the airfoil of my own design will have the highest surface area to volume ratio as it was designed following the shape of real airfoils, but its shape was arbitrarily determined without a specific shape in mind.

2. Mathematical background

For the purposes of this investigation, the objects investigated will be assumed to be geometrically symmetrical objects such that when revolved around an axis, the produced volume of revolution will be a perfect volume of revolution.

2.1) Volumes of revolution

The volume of revolution about the y-axis is given by the formula:

$$V = \int_a^b \pi f(x)^2 dx$$

The formula is derived from the Riemann sum, which allows the approximation of the area under a curve on a continuous function across the domain $x \in [a, b]$ by considering the sum of areas of regularly spaced rectangles under the curve. (Area of a Surface of Revolution | Calculus II) The sum is given by:

$$A = \sum_{i=1}^n \Delta x f(x_i)$$

One limitation of the Riemann sum is that when Δx is non-negligible, the Riemann sum can either overestimate or underestimate the total area under the curve. This problem is resolved by considering that as n approaches infinity, the width of each rectangle becomes infinitesimally small and from this, the Riemann integral is obtained. This allows the accurate calculation of the area under the curve for any continuous definite integral across the domain $x \in [a, b]$. The formula for the volume of revolutions can be derived from the consideration that the total area can be rotated 2π radians around the x-axis, therefore creating infinitely many cylinders of infinitesimal length with respective radius of $f(x_i)$. These cylinders are effectively circles in the 2D plane, and the volume of revolution can therefore be approximated simply by the sum of all the cross-sectional areas of these circles.

$$V \approx \sum_{i=1}^n \pi f(x_i)^2$$

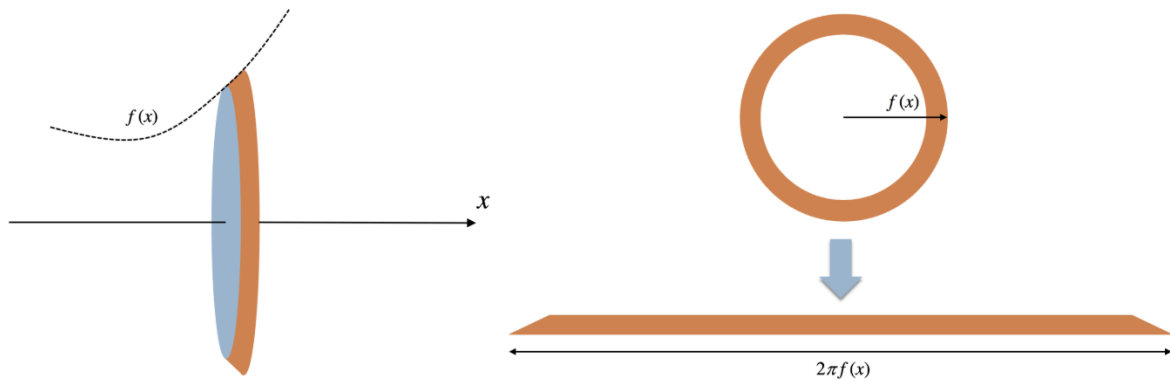
For all x across the domain $x \in [a, b]$, which when n approaches infinity, is equal to the integral:

$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi f(x_i)^2 = \int_a^b \pi f(x)^2 dx$$

Which is our equation for the volume of revolution. For the volume, the slope of the curve is not taken into account as the length of the cylinders are negligible as n approaches infinity.

2.2) Surface area of revolution

When considering the surface area of revolution, however, we can imagine an infinitesimally thin slice of a function $f(x)$, rotate it around the x-axis, which is equivalent in area to a trapezoid of length $2\pi f(x)$ (Tung).



(Fig 1. A visual representation of the surface area of an infinitesimally thin slice of $f(x)$)

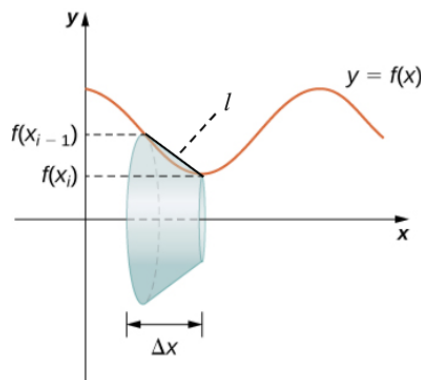
From this, we know that the surface area of each slice is dependent on the height of the trapezoid, which varies as a direction function of the gradient. Therefore, since the gradient is relevant, the curve must be represented by conical frustums instead of cylinders. A frustum is created by using regularly spaced trapezoids instead of rectangles to consider the area under the curve and similarly rotating it around the x-axis. As with the volume, it is important to have an infinite number of frustums to model the curve in order to produce an accurate value for the surface area. (Area of a Surface of Revolution | Calculus II)

For each frustum, the lateral surface area, A_f is calculated using the equation:

$$A_f = \pi(R + r)l$$

(Eq.1)

Where R is the radius of the lower base and r is the radius of the upper base, and l is the slant height.



Imagine a frustum, F , on the interval $[x_{i-1}, x_i]$ such that $\{[x_{i-1}, x_i]\} \subseteq \{[a, b]\}$, with lateral surface area A_l . As seen in (Fig. 2), $R = f(x_{i-1})$ and $r = f(x_i)$. We can use the pythagorean theorem to express l in terms of x and y in the form:

$$l^2 = \Delta x^2 + (\Delta y_i)^2$$

(Fig. 2 An image of a frustum)

Therefore,

$$l = \sqrt{\Delta x^2 + (\Delta y_i)^2}$$

Since l is a scalar distance, only the positive value is considered. Furthermore, by factoring out Δx^2 , we obtain:

$$l = \sqrt{\Delta x^2 (1 + (\frac{\Delta y_i}{\Delta x})^2)}$$

$$l = \Delta x \sqrt{1 + (\frac{\Delta y_i}{\Delta x})^2}$$

(Eq.2)

However, this expression for l is specific to F . To generalise this expression across the entire domain of the function, we must consider the mean value theorem. The mean value theorem states that “if f is continuous over the closed interval $[a, b]$ and differentiable over the open interval (a, b) , then there exists a point $c \in (a, b)$ such that the tangent line to the graph of f at c is parallel to the secant line connecting $(a, f(a))$ and $(b, f(b))$.” (4.2: The Mean Value

Theorem) Essentially, this theorem states that in F , the gradient of the tangent at some point x_i^* is equal to the gradient of the secant line $\frac{\Delta y_i}{\Delta x}$, or $f'(x_i^*) = \frac{\Delta y_i}{\Delta x}$, if and only if that if $f(x)$ is continuous and differentiable across the intervals $[x_{i-1}, x_i]$ and (x_{i-1}, x_i) respectively. Using this, we can alter (Eq.2) to the form:

$$l = \Delta x \sqrt{1 + (f'(x_i^*))^2}$$

Substituting this back into (Eq.1), we obtain:

$$A_f = \pi[f(x_{i-1}) + f(x_i)]\Delta x \sqrt{1 + (f'(x_i^*))^2} \quad (\text{Eq.3})$$

In order to express $f(x_{i-1})$ and $f(x_i)$ with a single variable, the intermediate value theorem must be considered. The intermediate value theorem states that “If f is continuous on the interval $[a, b]$ and N is between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$, then there is some number c in (a, b) such that $f(c) = N$.” (Continuity and IVT)

Therefore, there exists some point x_i^{**} such that:

$$f(x_i^{**}) = \frac{1}{2} (f(x_{i-1}) + f(x_i))$$

Therefore,

$$2f(x_i^{**}) = f(x_{i-1}) + f(x_i)$$

Substituting this back into (Eq.3), we get:

$$A_f = 2\pi f(x_i^{**})\Delta x \sqrt{1 + (f'(x_i^*))^2} \quad (\text{Eq.4})$$

The lateral surface area is therefore given by the sum of all the lateral surface areas of the infinitesimally wide frustums. This can be expressed as:

$$A_f \approx \sum_{i=1}^n 2\pi f(x_i^{**})\Delta x \sqrt{1 + (f'(x_i^*))^2}$$

This sum is similar to a Riemann sum with the exception of the evaluation of the function at x_i^{**} and x_i^* over the interval $[x_{i-1}, x_i]$. For a function $f(x)$ defined on $[a, b]$, we can divide the interval into n subintervals. Then, in each subinterval $[x_{i-1}, x_i]$ select x_i^* and x_i^{**} to evaluate the volume and surface area respectively. As $n \rightarrow \infty$, $\Delta x_i \rightarrow 0$, the difference between evaluating at x_i^* and x_i^{**} becomes negligible because the function is continuous and differentiable, therefore enabling the formula to simply use $f(x)$. (Strang and Herman)

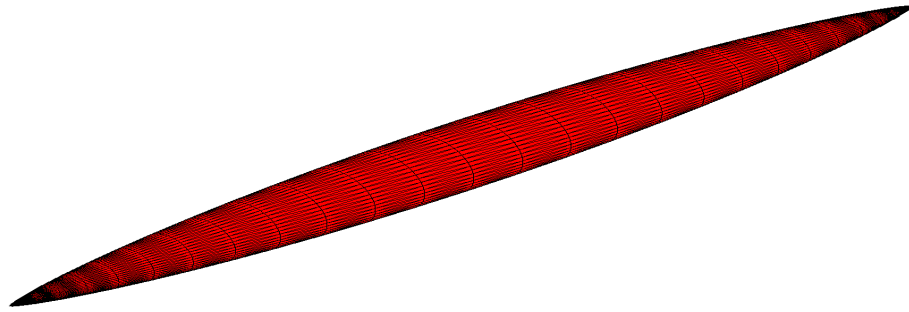
Therefore, it can be expressed as a definite integral in the form:

$$A_f = \lim_{n \rightarrow \infty} \sum_{i=1}^n 2\pi f(x_i^{**}) \Delta x \sqrt{1 + (f'(x_i^*))^2} = \int_a^b (2\pi f(x) \sqrt{1 + (f'(x))^2}) dx \quad (\text{Eq.5})$$

2.3) Sears-Haack Body

A Sears-Haack body is defined by the relationship (*Sears-Haack Body*)

$$r(x) = R_{max} [4x(1 - x)]^{\frac{3}{4}}$$



(Fig. 3 An example of a Sears-Haack Body)

Where $r(x)$ is the radius at a given x , R_{max} is the largest value of r which occurs at exactly the midpoint of the body, and $x \in [0, 1]$ where x is the fractional distance along the body such that $x = 0$ at the nose and $x = 1$ at the tail.

The cross sectional area S is given by:

$$S(x) = \pi r(x)^2 = \pi R_{max}^2 [4x(1 - x)]^{\frac{3}{2}}$$

$$S(x) = 8\pi R_{max}^2 (x)^{\frac{3}{2}} (1 - x)^{\frac{3}{2}}$$

To find volume V , integrate $S(x)$ from 0 to 1, since x is a fractional distance:

$$V = \int_0^1 S(x) dx = 8\pi R_{max}^2 \int_0^1 (x)^{\frac{3}{2}} (1 - x)^{\frac{3}{2}} dx$$

Since x is a fractional distance, we must multiply the integral by the length of the body, L , such that:

$$V = 8\pi R_{max}^2 L \int_0^1 (x)^{\frac{3}{2}} (1 - x)^{\frac{3}{2}} dx$$

(Eq.6)

Recognising that we can use trigonometric substitution to substitute $x = \sin^2 \theta$,

$$dx = 2 \sin \theta \cos \theta d\theta,$$

And when:

$$x = 0, \theta = 0$$

$$x = 1, \theta = \frac{\pi}{2}$$

And,

$$(1 - x) = \cos^2 \theta$$

The integral becomes:

$$\begin{aligned} \int_0^{\frac{\pi}{2}} (\sin^2 \theta)^{\frac{3}{2}} (\cos^2 \theta)^{\frac{3}{2}} 2 \sin \theta \cos \theta d\theta &= \int_0^{\frac{\pi}{2}} (\sin^3 \theta) (\cos^3 \theta) 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\frac{\pi}{2}} \sin^4 \theta \cos^4 \theta d\theta \end{aligned}$$

(Eq.7)

Since,

$$\sin(2\theta) = 2 \sin \theta \cos \theta,$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin(2\theta)$$

The integrand can therefore be simplified to:

$$\sin^4 \theta \cos^4 \theta = \left(\frac{1}{2}\sin(2\theta)\right)^4 = \frac{1}{16}\sin^4(2\theta)$$

(Eq.8)

The integral is now:

$$\frac{2}{16} \int_0^{\frac{\pi}{2}} \sin^4(2\theta) d\theta = \frac{1}{8} \int_0^{\frac{\pi}{2}} \sin^4(2\theta) d\theta$$

Substituting $\phi = 2\theta$ to further simplify,

$$d\phi = 2d\theta$$

The integral becomes:

$$\frac{1}{8} \times \frac{1}{2} \int_0^{\pi} \sin^4 \phi d\phi = \frac{1}{16} \int_0^{\pi} \sin^4 \phi d\phi$$

(Eq.9)

Using the cosine identity:

$$\begin{aligned} \cos(2\phi) &= 1 - 2\sin^2 \phi \\ \sin^2 \phi &= \frac{1 - \cos(2\phi)}{2} \end{aligned}$$

The integrand is:

$$\left(\frac{1 - \cos(2\phi)}{2}\right)^2 = \frac{1 - 2\cos(2\phi) + \cos^2(2\phi)}{4}$$

Integrating term by term we get:

$$\begin{aligned} \int_0^{\pi} \sin^4 \phi d\phi &= \frac{1}{4} \left[\int_0^{\pi} 1 d\phi - 2 \int_0^{\pi} \cos(2\phi) d\phi + \int_0^{\pi} \cos^2(2\phi) d\phi \right] \\ &= \frac{1}{4} \left(\pi + 0 + \frac{\pi}{2} \right) \end{aligned}$$

Since the first two integrals are solvable by inspection, the last integral can be evaluated as:

$$\cos(2\phi) = 2\cos^2 \phi - 1, \therefore \cos^2 \phi = \frac{1 + \cos(2\phi)}{2}$$

$$\begin{aligned} \int_0^{\pi} \cos^2(2\phi) d\phi &= \int_0^{\pi} \frac{1 + \cos(4\phi)}{2} d\phi \\ &= \frac{1}{2} \left[\int_0^{\pi} 1 d\phi + \int_0^{\pi} \cos(4\phi) d\phi \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{\pi}{2} + \frac{1}{2} \times \frac{1}{4} [\sin(4\phi)]_0^\pi \\
&= \frac{\pi}{2} + 0 = \frac{\pi}{2}
\end{aligned}$$

Therefore,

$$\int_0^\pi \sin^4 \phi \, d\phi = \frac{1}{4} \times \frac{3\pi}{2} = \frac{3\pi}{8}$$

Substituting back into (Eq.8) and (Eq.9),

$$\frac{3\pi}{8} \times \frac{1}{16} = \frac{3\pi}{128}$$

Finally, substituting into (Eq.6),

$$V = 8\pi R_{max}^2 L \left(\frac{3\pi}{128} \right)$$

$$V = \frac{3\pi^2}{16} R_{max}^2 L$$

L will be equal to 10 arbitrary units in this investigation, and will be kept constant when modelling all airfoils considered for the sake of simplicity. Therefore,

$$V = \frac{15\pi^2}{8} R_{max}^2$$

(Eq.10)

2.4) Defining R_{max}

Once the volume of revolution generated by my own airfoil has been calculated, R_{max} will be calculated by taking the average of the maximum points of all the other objects. This will allow for a volume to be calculated for the Sears-Haack body using (Eq.10). I considered fixing the volume of the body by equating it to the volume of another airfoil, but ultimately decided R_{max} would be fixed since the aim of this investigation is to compare the surface area to volume ratio for these bodies, it makes more sense to fix R_{max} rather than the surface area to volume ratio.

Also for this reason, I chose to use the average of the maxima of the other airfoils as, since R_{max} can take on any value, for any comparison to be reasonable the objects should be geometrically similar. As seen in (Fig 3), (Fig 4), and (Fig 7), the other airfoils are noticeably different to the Sears-Haack body as they are frontloaded in terms of their volume, whilst the Sears-Haack body's volume is evenly distributed along its length. Therefore, fixing R_{max} allows the maximum

radial height of the Sears-Haack body to be similar to the maximal radial height of the other airfoils, allowing for a fairer comparison.

2.5) Surface area of revolution for a Sears-Haack body

The surface area of revolution cannot be easily calculated as the resulting integrand is complex and difficult to integrate. For this reason, the surface area function will be derived for a Sears-Haack body and Wolfram Alpha will be used to calculate the approximate surface area of the body using numerical integration. The usage of polynomial regression to approximate the curve of the Sears-Haack body was considered, but a polynomial cannot accurately model the behaviour of a Sears-Haack body near the endpoints of the curve, so the following function will still be used to calculate the surface area of the body.

To use (Eq.5), we need to first find the derivative of $r(x)$.

Calculating $\frac{dr}{dx}$,

$$r(x) = R_{max} [4x(1-x)]^{\frac{3}{4}}$$

$$\frac{dr}{dx} = 3R_{max} \frac{1-2x}{(4x(1-x))^{\frac{1}{4}}}$$

$$\left(\frac{dr}{dx}\right)^2 = \left(\frac{3R_{max}}{L}\right)^2 \left(\frac{1-2x}{(4x(1-x))^{\frac{1}{4}}}\right)^2$$

Substituting into (Eq.5), the surface area of a Sears-Haack body, A_{sh} , is expressed as:

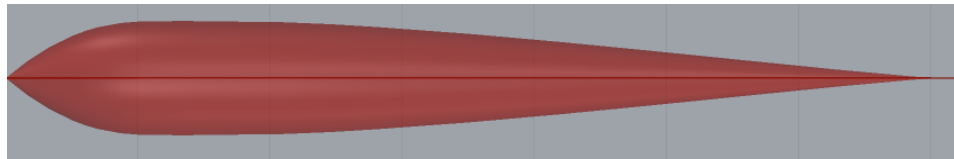
$$A_{sh} = 2\pi R_{max} L \int_0^1 [4x(1-x)]^{\frac{3}{4}} \sqrt{1 + \frac{9R_{max}^2}{L^2} \times \frac{(1-2x)^2}{\sqrt{(4x(1-x))}}} dx \quad (\text{Eq.11})$$

2.5) Graphing the airfoils

2.5.1) My airfoil design

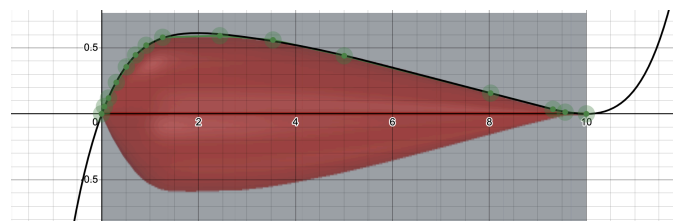
Once the formula for the surface area and volume of the objects were derived, I moved onto graphing the upper curve for each of the objects. The following volume of revolution was

obtained through the modelling software Rhino8 and coded in Grasshopper, and the parent function of the curve is a bézier curve. A bézier curve is a curve that can be manually changed to suit design needs by dragging various control points which create the curvature of the function. Details can be found in Appendix B. For the purposes of this investigation, this curve will simply be approximated using polynomial regression through Desmos graphing calculator at <https://www.desmos.com/calculator>, as the determination of elementary functions that can represent this bézier curve accurately is beyond the scope of this investigation, and the increase in accuracy will be negligible.



(Fig 4. My airfoil design)

I considered modelling this curve through piecewise functions, using the natural log function when the graph is increasing, as the slope significantly decreases in the sections of the graph closest to the origin. I also considered using interpolation, specifically for the next airfoil, as I had access to coordinate data for it (Appendix A), and this would not be subjected to human error. However, due to the size of the data set, methods such as Lagrangian interpolation would be inaccurate as the imported data contained 68 points. Therefore, I decided to try to model the object through polynomial regression.



(Fig 5. Polynomial regression model for my airfoil design)

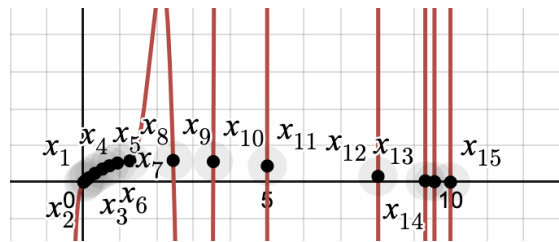
Note: this image appears different from the one above as the viewing boundaries in Desmos have been shifted. The image imported into Desmos is exactly as seen above.

This object model was generated using Rhino8 and Grasshopper by creating a curve which was then revolved about the x-axis. A screenshot was then taken of the produced solid of revolution and imported into Desmos graphing calculator.

This regression model was generated using 15 manually plotted data points, and the shown function is a 7th degree polynomial

$$f(x) = 0.00000746029x^7 - 0.000296076x^6 + 0.00485066x^5 - 0.0424024x^4 + 0.214093x^3 - 0.636079x^2 + 0.99501x - 0.00491353$$

A 7th degree polynomial was used to reach $R^2 > 0.99$ such that the object was modelled as accurately as possible. Furthermore, the decision to use polynomial regression instead of Lagrangian interpolation was justified, as when using an online calculator via Desmos, the following graph was generated.



(Fig 6. Approximation of my airfoil design generated by Lagrangian interpolation)

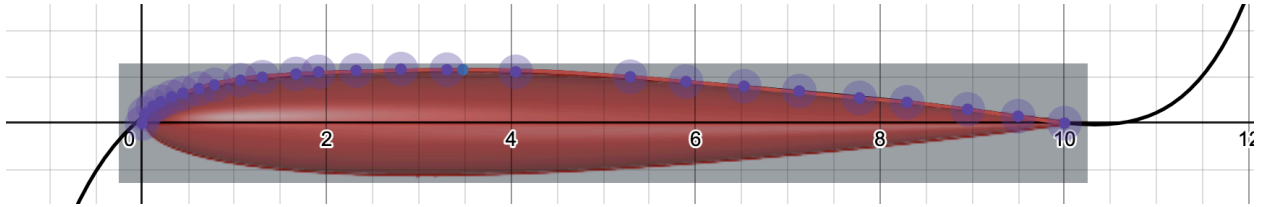
This is a result of Runge's phenomenon (*Explore Runge's Polynomial Interpolation Phenomenon*), where behaviour of the graph oscillates wildly near the endpoints. Finally, I did not pursue a higher degree of accuracy by including terms with higher powers of x , as this has potential to overfit the data.

2.5.2) NACA0012 airfoil

This object was modelled using imported coordinate data (n0012-II).



(Fig 7. NACA0012 Airfoil)



(Fig 8. Polynomial regression model for NACA0012 airfoil)

The shown function was calculated using 25 manual points within Desmos as

$$g(x) = 0.0000113206x^7 - 0.000434487x^6 + 0.00670712x^5 - 0.0533706x^4 \\ + 0.234366x^3 - 0.581703x^2 + 0.802121x + 0.0618976$$

The R^2 value for this curve is 0.9909. The image itself was generated within Rhino8 by inputting the coordinate data into the modelling software to generate a curve which was then revolved about the x-axis to generate the solid of revolution. I did not plot the 68 given data points as there was no simple way to input this data into Desmos. Instead, the airfoil was accurately modelled within Rhino8 using Grasshopper, found in Appendix C, and imported into Desmos graphing calculator as a screenshot.

2.6) Defining R_{max}

To generate a value for R_{max} from the average of the maxima for $f(x)$ and $g(x)$, I considered setting the derivative of each function to zero and solving, but a far more efficient approach was to use Desmos to provide the exact coordinates for each maxima. The maxima are as follows

$$f(x)_{max} = (1.97387, 0.61236)$$

$$g(x)_{max} = (3.47599, 0.57983)$$

The value of R_{max} is given by

$$\frac{(0.61236+0.57983)}{2} = 0.59610 \text{ rounded to 5 decimal places.}$$

2.7) Calculating volumes of revolution and associated surface areas

2.7.1) My airfoil design

The volume is calculated using

$$V = \int_0^{10} \pi f(x)^2 dx = 5.29663 \text{ units}^3$$

The volume is calculated using a TI-nspire CX II graphical display calculator as expanding $f(x)^2$ by hand is unnecessarily tedious.

The surface area is calculated first by finding the derivative of $f(x)$ using the power rule.

$$\begin{aligned} f'(x) = & 0.00005222203x^6 - 0.001776456x^5 + 0.0242533x^4 - 0.1696096x^3 \\ & + 0.642279x^2 - 1.272158x + 0.99501 \end{aligned}$$

The surface area is then

$$\int_0^{10} (2\pi f(x) \sqrt{1 + (f'(x))^2}) dx = 22.8317 \text{ units}^2$$

The surface area to volume ratio is

$$\frac{22.8317}{5.29663} = 4.31061 \text{ to 5 decimal places}$$

2.7.2) NACA0012 curve

As above, the volume is calculated using

$$V = \int_0^{10} \pi g(x)^2 dx = 5.90057 \text{ units}^3$$

The surface area is calculated first by finding the derivative of $g(x)$ using the power rule.

$$\begin{aligned} g'(x) = & 0.0000792442x^6 - 0.002606922x^5 + 0.0335356x^4 - 0.2134824x^3 \\ & + 0.703098x^2 - 1.163406x + 0.802121 \end{aligned}$$

The surface area is then

$$\int_0^{10} (2\pi g(x) \sqrt{1 + (g'(x))^2}) dx = 25.2744 \text{ units}^2$$

The surface area to volume ratio is

$$\frac{25.2744}{5.90057} = 4.28338 \text{ to 5 decimal places}$$

2.7.3) Sears-Haack body

The formula for the volume has been derived as (Eq.11), where $R_{max} = 0.59610$

$$V = \frac{15\pi^2}{8} \times 0.59610^2 = 6.57565 \text{ to 5 decimal places}$$

The surface area is calculated by inputting the following function into Wolfram Alpha at <https://www.wolframalpha.com/>.

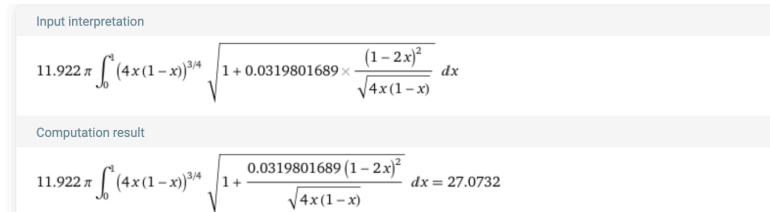
$$A = 11.922\pi \int_0^1 [4x(1-x)]^{\frac{3}{4}} \sqrt{1 + 0.0319801689 \times \frac{(1-2x)^2}{\sqrt{4x(1-x)}}} dx$$

$$A \approx 27.0732 \text{ units}^2 \text{ to 4 decimal places}$$

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11.922pi * the numerical integration of (4x(1-x))^(3/4) * sqrt((1+0.0319801689*(1-2x)^2/sqrt(4x*(1-x))))



(Fig 9. Numerical integration calculation via Wolfram Alpha)

It is important to note that this is an approximation as the calculation of the surface area for the Sears-Haack body utilised numerical integration on Wolfram Alpha, which is an approximation technique, as the definite integral is not easily integrable analytically.

The surface area to volume ratio is

$$\frac{27.0732}{6.57565} = 4.1172 \text{ to 4 decimal places}$$

2.8) Tabulated results

Object	Surface Area (Units ²)	Percentage difference with Sears-Haack surface area	Volume (Units ³)	Percentage difference with Sears-Haack volume	SA:V ratio to 4 decimal places	Percentage difference with Sears-Haack SA:V ratio
Sears-Haack Body	27.0732		6.57565		4.1172	
NACA0012 airfoil	25.2744	-6.6442%	5.90057	-10.2664%	4.2834	+4.0367%
My airfoil	22.8317	-15.6668%	5.29663	-19.4509%	4.3106	+4.6974%

(Table 1. A table comparing the surface area, volume and SA:V ratio of all 3 objects)

The percentage difference was calculated using the formula

$$\% \text{ difference} = \frac{\text{Compared value} - \text{Original value}}{\text{Original value}} \times 100$$

The following calculation was done for the percentage difference with Sears-Haack SA:V ratio for the NACA0012 airfoil.

$$\frac{4.2834 - 4.1172}{4.1172} \times 100 = 4.0367\%$$

3. Conclusion

This investigation aimed to compare the surface area to volume ratio of various different objects in an aeronautical context. In order to achieve this aim, I utilised my knowledge of calculus in order to derive expressions for the surface area of solids of revolutions, as well as a general volume and surface area expression for the Sears-Haack body. After this, I modelled all the objects investigated within appropriate software and plotted points on graphs in order to derive the functions for these curves. Using these functions, I used my derived formulas to calculate the

surface areas and volumes for the solids of revolution. Comparing my results, it is clear that my hypothesis that the Sears-Haack body would have the lowest SA:V ratio is true. It should be noted that although the ratios between the investigated objects are relatively close in value, because this Sears-Haack body was generated using geometric data from the other objects, it is not the most optimised form of the body.

It is also important to note that the Sears-Haack body has a significantly higher volume than the other two objects. This can easily be inferred from the fact that the Sears-Haack body has a uniform volume distribution around $x=0.5$, whereas the other two objects are more front heavy, meaning that a large portion of their volumes are distributed in the front of the object. Since this front heavy section contains the maximum radial length from the axis of symmetry for both objects, by generating R_{max} from these values, it is expected that the Sears-Haack body has a higher volume. The surface area of the Sears-Haack body is also larger than those of the other objects, which can be expected as it encompasses a larger volume.

Beyond this, I also familiarised myself with graphical approximation methods that are accurate to a high degree, which may be useful in further applications such as modelling other objects, as well as finding ways to determine if an object can be analysed in the manner presented in this investigation, which will be discussed in the evaluation.

4. Evaluation

The strengths of my investigation include high degrees of accuracy, such as by using industrial data as well as using R^2 values to determine accuracy, within the modelling of the objects minimised the uncertainty from human error. This increased the accuracy and reliability of the results. The choice of tools utilised also benefited this investigation, as by using a simpler approach the results remained highly accurate whilst avoiding any potential issues such as Runge's phenomenon. Additionally, the integral presented in (Eq.6) could have been evaluated using the Beta and Gamma functions, but the explanation of these was beyond the scope of this investigation. Therefore, the more tedious trigonometric substitution approach seen in this investigation was used.

Through the investigation, I was also able to determine that due to the mean and intermediate value theorems, objects that could not be represented as continuous and differentiable over the intervals $[a, b]$ and (a, b) respectively could not be analysed as in this investigation. These objects include those with sharp corners or discontinuities.

However, this investigation was also subjected to some limitations. Firstly, by manually plotting points on Desmos to approximate the function of the curve, the curve was subjected to human error in visually interpreting the image. Secondly, although there was no easy way to import large amounts of data into Desmos, possible alternatives include writing code that could interpret and generate a polynomial regression model based on coordinate data, which would be both accurate and time efficient and could further improve the accuracy of the results.

Finally, the airfoil of my design had a surface area to volume ratio within 1% of the NACA0012 airfoil, which was satisfactory due to the lack of planning surrounding the design of the airfoil. After further optimisation, I expect the surface area to volume ratio to further decrease, hopefully below the ratio of the NACA0012 airfoil. This optimisation can be done by distributing the volume more evenly across the airfoil. It is important to note that traditional theories of wing design cannot be taken into account for this investigation, specifically the idea of cambered wings as the airfoil design must retain axial symmetry to be a valid object for volumes of revolutions.

To further develop this investigation, the alternative approach presented in section 2.4 could be used, that is fixing the volume of the Sears-Haack body to the average of the volumes of the other objects. This approach would allow the determination of the most aerodynamic shape for a given length and volume. The results from this approach could then be compared to the results of this investigation to find out whether it is more efficient in terms of surface area to volume ratio to fix the volume or the maximal radial distance of a Sears-Haack body.

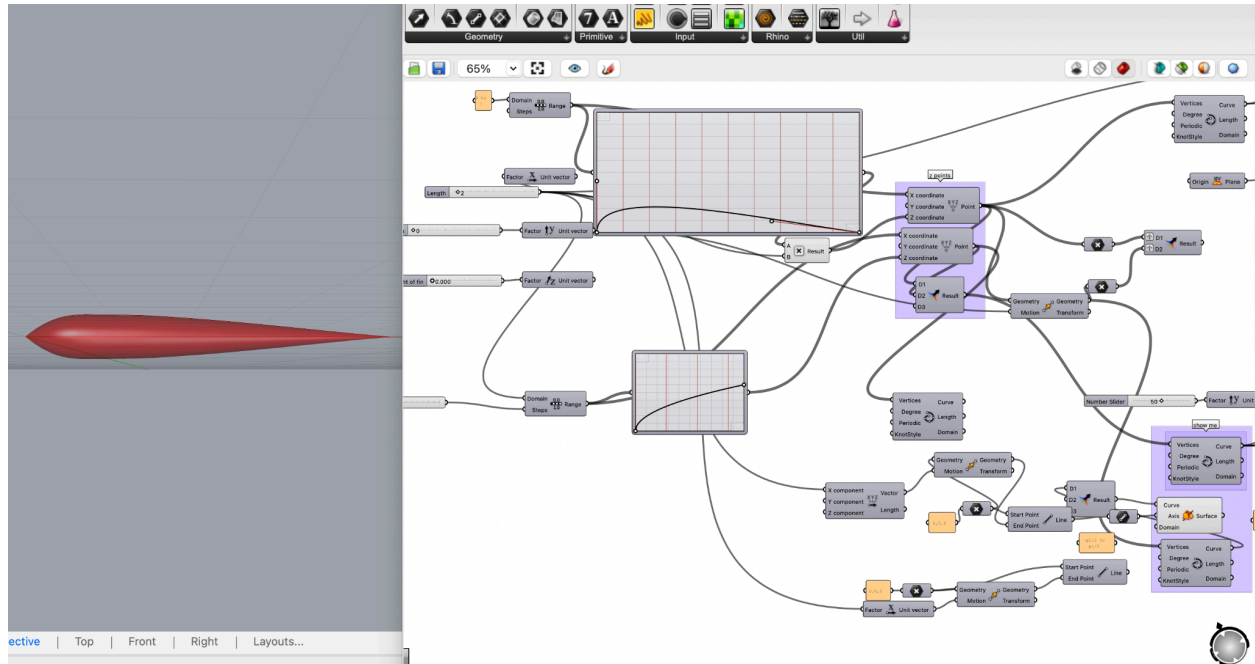
Appendix A - NACA0012 airfoil coordinate data

x	y
0	0
0.0005839	0.0042603
0.0023342	0.0084289
0.0052468	0.0125011
0.0093149	0.0164706
0.0145291	0.02033
0.0208771	0.0240706
0.0283441	0.0276827
0.0369127	0.0311559
0.0465628	0.0344792
0.057272	0.0376414
0.0690152	0.040631
0.0817649	0.0434371
0.0954915	0.0460489
0.1101628	0.0484567
0.1257446	0.0506513
0.1422005	0.0526251
0.1594921	0.0543715
0.1775789	0.0558856
0.1964187	0.057164
0.2159676	0.0582048
0.2361799	0.0590081

0.2570083	0.0595755
0.2784042	0.0599102
0.3003177	0.0600172
0.3226976	0.0599028
0.3454915	0.0595747
0.3686463	0.0590419
0.3921079	0.0583145
0.4158215	0.0574033
0.4397317	0.05632
0.4637826	0.0550769
0.4879181	0.0536866
0.5120819	0.052162
0.5362174	0.0505161
0.5602683	0.0487619
0.5841786	0.0469124
0.6078921	0.0449802
0.6313537	0.0429778
0.6545085	0.0409174
0.6773025	0.0388109
0.6996823	0.03667
0.7215958	0.0345058
0.7429917	0.0323294
0.7638202	0.0301515
0.7840324	0.0279828

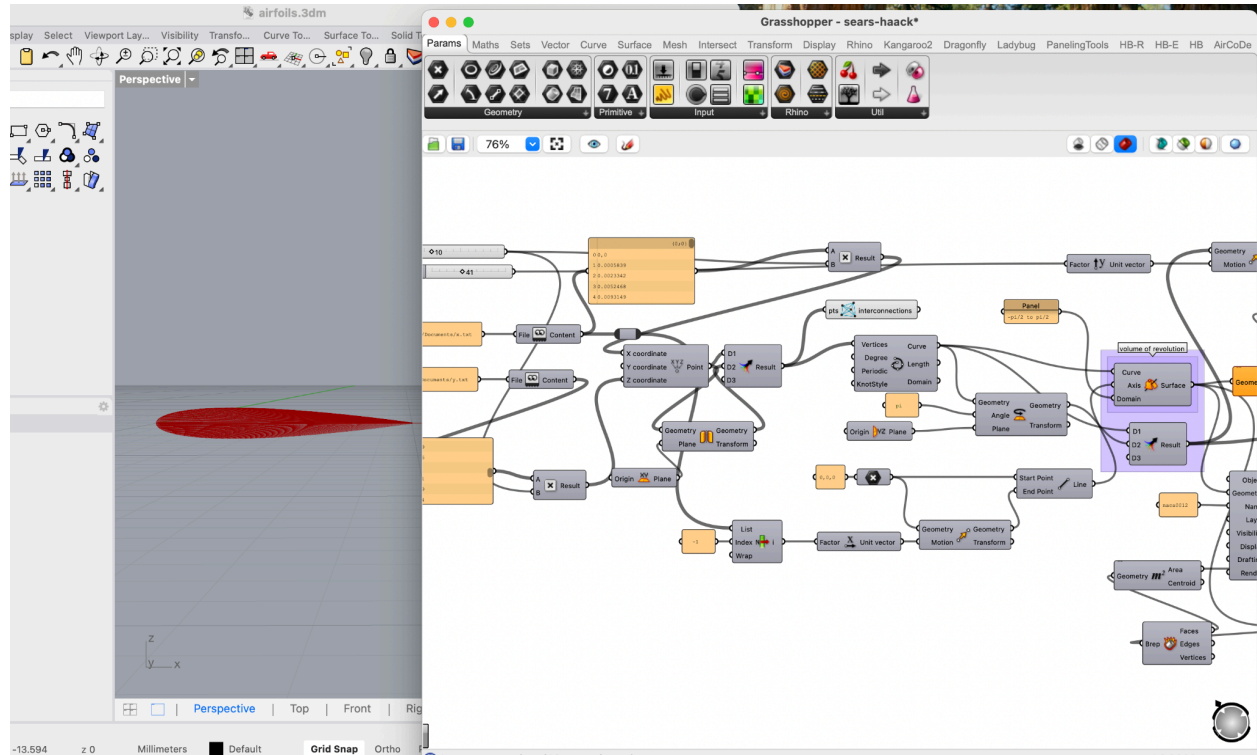
0.8035813	0.0258337
0.8224211	0.0237142
0.8405079	0.0216347
0.8577995	0.0196051
0.8742554	0.0176353
0.8898372	0.0157351
0.9045085	0.0139143
0.9182351	0.0121823
0.9309849	0.0105485
0.942728	0.0090217
0.9534372	0.0076108
0.9630873	0.0063238
0.9716559	0.0051685
0.9791229	0.0041519
0.9854709	0.0032804
0.990685	0.0025595
0.9947532	0.0019938
0.9976658	0.001587
0.9994161	0.0013419
1	0.00126

Appendix B - Modelling my airfoil design in Rhino8 and Grasshopper



Note. The following object was modelled using parametric controls through Grasshopper coding done through the Rhino8 software.

Appendix C - NACA0012 airfoil modelling in Rhino8 and Grasshopper



Note. The following object was modelled by importing the coordinate data found in Appendix A into Grasshopper, then revolving the coordinate data by 2π in Grasshopper code.

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